# Technical Writeup - MAM-AI

*Gemma 3n Impact Challenge*




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This submission is on behalf of

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## EXECUTIVE SUMMARY

Zanzibar experiences unacceptably high rates of maternal and newborn mortality, primarily due to lack of information at the point of care. MAM-AI is an offline-first, privacy-preserving, clinical smart-search app designed to support nurses and midwives in Zanzibar for neonatal care. The key challenge of these nurses is information scarcity - doctors and supervisors are often far away and unavailable. MAM-AI addresses this by putting easy summaries of gold-standard clinical guidelines right in the hands of the nurses in need at the point of care. Nurses type their queries into the search bar, the app searches through the guidelines, and they receive a summary of the relevant clinical information. MAM-AI aims to reduce preventable maternal and neonatal deaths by democratizing access to expert medical information at the point of care.

## INTRODUCTION & PROBLEM DESCRIPTION

**Every two minutes, a woman dies from preventable complications of pregnancy or childbirth**, totalling more than 260,000 lives lost each year. Sub-Saharan Africa bears a disproportionate share of this crisis, accounting for roughly two-thirds of all maternal deaths worldwide. Most of these deaths occur during delivery or within the first 48 hours postpartum, an interval when prompt intervention can save lives.

Yet in many facilities, even when care is available, providers lack the timely, evidence based information needed to guide life-saving decisions. Studies across sub-Saharan Africa's primary healthcare settings reveal consistent barriers in accessing clinical decision-support tools, with technical and infrastructural limitations preventing the translation of available medical knowledge into practice [1, 2]. This creates a critical "know-do gap" where life-saving protocols exist but remain inaccessible at the point of care.

Zanzibar exemplifies this unacceptable reality: maternal deaths occur at rates 32% higher and newborn deaths at rates 70% higher than the global average [3, 4, 5]. Our October 2024 field assessment, conducted (by team member **Trevor Brokowski**) across Zanzibar's three levels of care, engaging with the Zanzibar Ministry of Health, the Zanzibar Nursing Association, and healthcare providers from primary to tertiary facilities, confirmed that **information poverty** is central to this crisis.

Consider the reality for a midwife confronting postpartum hemorrhage in a rural dispensary. Though the emergency is recognized, the facility's clinical handbook, the resource most commonly used to advise decision making, only offers generic protocols. Their supervisor is unreachable due to poor network coverage, and data costs put Google searches out of reach. With the nearest referral hospital hours away, they must rely solely on their (potentially limited) training.

This scenario is not an exception but the daily reality for nurses and midwives across Zanzibar's healthcare system. **Nurses and midwives must practice life-saving medicine in an information desert when immediate access to evidence-based guidelines could transform outcomes.**

### The MAM-AI Solution

The MAM-AI initiative addresses these issues through an innovative offline AI-powered search application purpose-built for Zanzibar's healthcare realities, placing evidence-based maternal care guidance directly into the hands of nurses and midwives.

At its core, MAM-AI integrates the Gemma-3n language model with Retrieval Augmented Generation (**RAG**) on an **expert-curated guideline repository** to deliver accurate, **context-specific responses** to maternal care queries. When a provider searches for clinical direction, the system instantly retrieves the most relevant guidance from a compact, 11 MB database of medical protocols. Once deployed, MAM-AI operates **entirely offline**, removing reliance on unstable connectivity that has historically undermined technology adoption in similar settings.

MAM-AI represents a practical pathway to healthcare equity, ensuring that every provider, regardless of location or infrastructure constraints, can access the clinical guidance needed to deliver timely, evidence-based care, reducing preventable maternal and newborn deaths in Zanzibar.

## METHODS

We made use of Google's AI Edge RAG for Android SDK for our MAM-AI solution.

Our objective was to fine-tune the Gemma-3n model and integrate it with RAG (Retrieval Augmented Generation) for enhanced context retrieval and response generation. While we successfully completed the fine-tuning process, we encountered challenges when converting the fine-tuned model to the .tflite format required for Android implementation. Due to time constraints, we were unable to resolve the unexpected bugs that arose during the deployment process. **Consequently, the fine-tuned model is not included in our final product submission - only the RAG model is.**

**The following discussion of our fine-tuning methodology and results is presented for completeness and documentation purposes**. This work represents a critical next step in our development roadmap. We strongly hypothesize that integrating the fine-tuned model with RAG will improve the response quality of the MAM-AI system.

### Datasets

#### RAG Context Dataset

For document retrieval and context generation, we compiled a collection of eight medical guideline documents for maternal care in PDF format, sourced by medical experts. These eight were drawn from a wider pool of documents judged by clinicians to be relevant to

antenatal and neonatal care. This curated collection spans approximately 650 pages of content and forms the foundation of our retrieval-augmented generation (RAG) system. We strategically prioritized documents focused on post-birth care over antenatal care to align with our application's primary use cases. These documents serve as the knowledge base for our RAG implementation, providing contextually relevant information that is dynamically retrieved based on user search queries. The document guidelines can be found here.

**Fine-tuning (MIRIAD) Dataset**

For fine-tuning the Gemma 3n model, we employed the MIRIAD dataset. “MIRIAD is a large-scale, curated corpus of 5,821,948 medical instruction–response pairs, each grounded in peer-reviewed literature” [6]. Due to project time and resource constraints, we utilized a subset of this dataset for our training purposes. The dataset was filtered as follows:

**Specialty breakdown:**

- **General Pediatrics:** 189,405
- **Family Medicine & Primary care:** 182,405
- **Obstetrics & Gynecology:** 130,263
- **Pediatric Subspecialties:** 324

**Total records filtered and saved: 502,189**

The filtered dataset was randomised and further reduced to one tenth the original size. It was then used to finetune the gemma-3n-E4B-it with Unsloth.

## Model and Fine-tuning - Gemma-3n

We used the Gemma-3n language model, in accordance with the Kaggle competition requirements. Specifically, we fine-tuned the gemma-3n-E4B-it variant provided by Unsloth, leveraging LoRA (Low-Rank Adaptation) with 4-bit quantization to optimize for efficiency on constrained hardware. The resulting model was tailored for medical question-answering tasks, with a focus on domain-specific accuracy and responsiveness.

For the initial prototype, only the RAG-based system underwent expert testing and deployment. However, the LoRA-fine-tuned Gemma model has been fully trained and is publicly available (link). The next phase of this project will involve expert validation of the fine-tuned model and conversion for deployment on Android devices.

## Base Model Specifications

The system utilizes the unsloth/gemma-3n-E4B-it model as the foundation for fine-tuning. This model provides:

- Architecture: Gemma-3n with instruction-tuning capabilities
- Sequence Length: 1024 tokens maximum processing capacity

- Quantization: 4-bit precision for memory efficiency
- Framework: Unsloth optimization framework for accelerated training

### LoRA Configuration Parameters

The fine-tuning process implements Low-Rank Adaptation with the following specifications:

| Parameter | Value | Purpose |
|---|---|---|
| LoRA Rank (r) | 16 | Determines adaptation capacity and parameter efficiency |
| LoRA Alpha | 16 | Scaling factor controlling adaptation strength |
| Target Modules | q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj | Attention and MLP layer targeting |
| Dropout Rate | 0.0 | Disabled for optimal convergence |
| Bias Adaptation | None | No bias parameter updates |

## Training Methodology

### Data Processing Pipeline

The training data undergoes structured preprocessing to ensure optimal model performance:

- Data Validation: Automatic removal of samples with missing questions or answers
- Format Standardization: Conversion to chat-based instruction format
- Template Application: Implementation of user-assistant conversation structure

Conversation Format Template:

```
<|user|>
{medical_question}
<|assistant|>
{medical_answer}<|end_of_text|>
```

## Training Configuration

The fine-tuning process employs the following training parameters:

- Batch Size: 1 (optimized for memory constraints)
- Training Epochs: 3 complete dataset iterations
- Learning Rate: 2e-4 with linear decay schedule



- Optimizer: AdamW with 8-bit precision
- Weight Decay: 0.01 for regularization
- Mixed Precision: Automatic FP16/BF16 selection based on hardware support

**Memory Optimization Strategies**

The implementation incorporates several memory efficiency techniques:

- 4-bit Model Quantization: Reduces memory footprint by approximately 75%
- Gradient Checkpointing: Trades computation for memory during backpropagation
- LoRA Parameter Efficiency: Minimizes trainable parameters while maintaining performance
- Unsloth Framework: Optimized kernels for faster training and reduced memory usage

## Implementation Components

**Configuration Module (config.py)**
Centralizes model settings, training parameters, and file paths including model name, sequence length, LoRA parameters, batch size, learning rate, and data directories.

**Data Processing Module (data_processing.py)**
Handles CSV data loading, missing value removal, conversation format conversion, and dataset preparation for training and evaluation.

**Model Setup Module (model_setup.py)**
Initializes the Gemma-3n model with 4-bit quantization, configures LoRA adapters for target modules, and enables gradient checkpointing for memory efficiency.

**Training Module (training.py)**
Implements the fine-tuning process using SFTTrainer with mixed precision training, AdamW optimizer, and automatic model checkpointing.

**Evaluation Module (evaluation.py)**
Provides model inference capabilities, answer generation with configurable parameters, automated evaluation on test datasets, and interactive testing interface.

**Deployment Process**

The fine-tuning workflow follows these steps:

- Configuration Setup: Define model parameters and training settings
- Data Preparation: Load and preprocess medical Q&A datasets
- Model Initialization: Setup Gemma-3n with LoRA adapters
- Training Execution: Fine-tune model using memory-optimized techniques
- Performance Evaluation: Test accuracy and response quality
- Model Deployment: Save trained model

## RAG (Retrieval Augmented Generation)

Our RAG implementation uses Google's AI Edge SDK for Android. We use the on-device

Gecko embedder model to embed documents in vector space for semantic retrieval.

For our RAG process, we utilized a medical guideline dataset consisting of PDF documents. To extract text from these PDFs, we employed mmore, a multimodal text extraction tool developed by LiGHT (team institution). Following text extraction, we used LangChain to split the text and optimize chunk sizes for retrieval performance.

Our text splitting approach employed a hybrid strategy using LangChain's built-in functions. We first split documents based on markdown format headings (levels 1, 2, and 3), then applied recursive text splitting with a fixed chunk size of 1000 characters. Due to hardware constraints, we found that 1000-character chunks with 200-character overlaps provided optimal performance. We developed a preprocessing program that chunks the text, adds SEP tokens as separators, and saves the output to text files for input into the AI Edge RAG Android SDK.

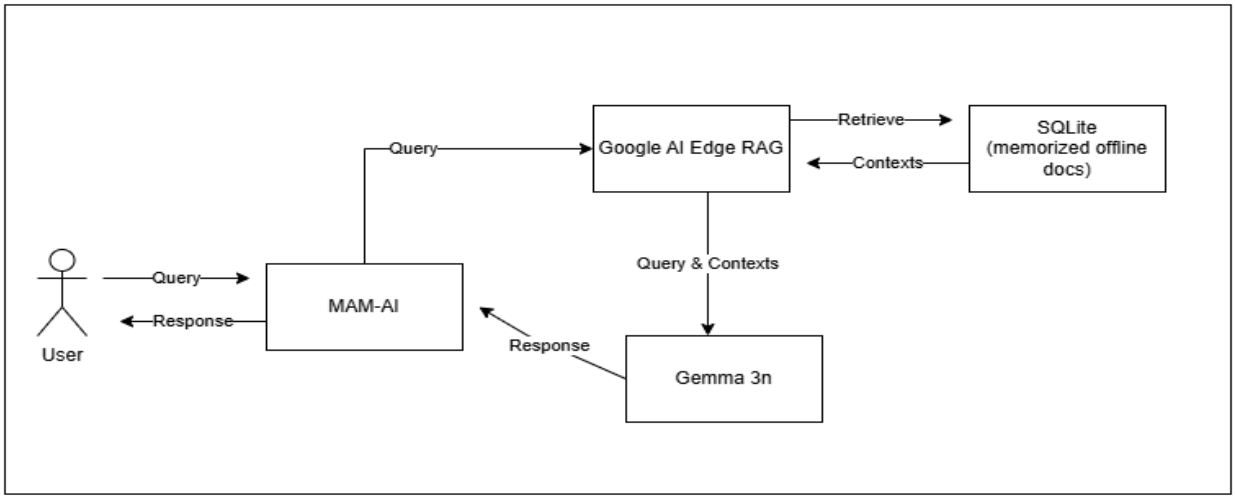
All processed chunks are stored in a local, offline SQLite database on the device, enabling retrieval without internet connectivity. When a user submits a query, the AI Edge SDK processes the request and performs retrieval from the local SQLite document store. The system retrieves contexts based on vector similarity score, selecting only the top 3 most relevant chunks from the database. The retrieved contexts are then passed alongside the original query to the Gemma-3n model, which generates a comprehensive response. To ensure transparency and interpretability, the system returns both the generated response and the retrieved contexts to the user, allowing them to understand the sources behind the AI's answer.

## Architecture of the MAM-AI app

The MAM-AI application is engineered as a fully offline, on-device system designed to deliver reliable clinical decision support on standard Android smartphones and tablets, specifically addressing the challenge of poor internet connectivity in Zanzibar. Our architecture prioritizes **clinical safety, speed, and accuracy**, ensuring that frontline healthcare workers have immediate access to evidence-based information at the point of care.

The application was built with **Flutter**, while the core logic is powered by Google's **AI Edge RAG for Android SDK**.

Testing was conducted on a Lenovo Yoga Tab Plus tablet running Android 14, equipped with a Qualcomm Snapdragon 8 Gen 3 processor and 16GB of RAM, providing robust hardware capabilities for evaluating the application's performance under resource-constrained conditions.




*A high-level overview of the MAM-AI on-device architecture, illustrating the flow from user input to the generation of a response.*

## Data Flow and Application Logic

The application's workflow is streamlined into a single screen to ensure simplicity and rapid access to information in high-pressure clinical settings.

1. **Initialization:** When the nurse launches the app, the system initializes two core components stored locally on the device: the **vector database** (managed by SQLite) containing the embedded medical guidelines, and the **Gemma 3n model** (packaged as a **.task** file).
2. **Query Input:** The user enters a clinical query (e.g., "management of postpartum hemorrhage") into the search bar presented by the Flutter UI.
3. **Retrieval:** The query string is passed to the **AI Edge RAG SDK**. The SDK uses the Gecko 110m embedding model to convert the query into a vector representation. It then performs a similarity search against the local SQLite vector database to find the most relevant text chunks from the source medical documents. The **top three most relevant chunks** with the **highest similarity score** are retrieved.
4. **Generation:** The three retrieved text chunks are combined with the original user query to form a detailed, context-rich prompt. This prompt is then passed to the Gemma 3n model.
5. **Response Display:** The Gemma model processes the prompt and generates a concise, synthesized answer grounded exclusively in the provided context. The Flutter UI then displays this final answer to the user. For transparency and to allow for further clinical verification, the three source text chunks used for generation are also displayed alongside the answer.

## Key Technical Components

### Frontend: Flutter

The user interface is developed using Flutter, Google's cross-platform UI toolkit. This



choice was made for two main reasons:

- **Rapid Development:** Flutter's declarative UI and hot-reload capabilities allowed for fast iteration on the user interface.
- **Familiarity:** the team was already familiar with Flutter, having each individually built several apps using the technology.

**Backend: On-Device Vector Database (SQLite)**

The core knowledge base of MAM-AI is a vector database managed within an on-device **SQLite** database. This database stores the vector embeddings of our curated medical guidelines.

- **Data Preparation:** The source PDF documents were first processed to extract clean text. This text was then divided into overlapping chunks to preserve clinical context.
- **Embedding and Storage:** Each text chunk was converted into a vector embedding and stored in the SQLite database. This entire process is done offline, and the final database file is bundled with the application, ensuring zero reliance on external servers.

**Core Logic: AI Edge RAG for Android SDK**

This SDK acts as the central nervous system of the application, orchestrating the entire Retrieval-Augmented Generation pipeline on the device. We leverage it for two critical functions:

1. **Query Embedding:** Converting the user's live text query into a vector for searching.
2. **Context Retrieval:** Efficiently querying the SQLite vector database to fetch the most relevant document chunks. By using this specialized SDK, we were able to implement a complex RAG pipeline optimized for mobile hardware without building the underlying infrastructure from scratch.

## Challenges

The challenges we faced during development include:

- **Model conversion issues**: Despite successfully fine-tuning the Gemma-3n model, we encountered unexpected bugs and conversion errors when attempting to convert to .tflite format for Android deployment, preventing integration of the fine-tuned model in our final submission.
- **Beta technology dependencies**: Relying on Google's AI Edge RAG SDK in beta status introduced unpredictability, and changes that required additional development time and workarounds
- **Performance optimization constraints**: The Gecko embedding model's slow inference speed required precomputing all document embeddings offline and extensive pipeline optimization to maintain acceptable response times
- **Hardware distribution challenges**: Ensuring healthcare providers have access to

suitable Android devices capable of running the application effectively, particularly in rural facilities with limited technology resources. We plan to address this during deployment by purchasing tablets for the clinicians.
- **Complex data processing**: Converting medical guideline PDFs into optimized retrieval formats while preserving clinical context and accuracy required custom preprocessing pipelines and extensive testing
- **System integration and debugging**: Resolving numerous technical issues including memory optimization problems, user interface inconsistencies, and maintaining clinical safety standards throughout development

### Limitations

The notable limitations of the MAM-AI system include:

- **Single-query architecture**: Functions as a search app rather than a conversational chatbot, preventing multi-turn conversations or follow-up questions within the same clinical context
- **No session memory**: Does not maintain conversation history or context between queries, requiring standalone queries rather than building upon previous responses
- **Static knowledge base**: Limited to pre-loaded medical guidelines with no ability to access updated protocols or new research findings without manual application updates
- **Hardware dependencies**: System performance constrained by Android device specifications, with potential slower response times on lower-end smartphones common in resource-limited settings
- **English-only interface**: Operates exclusively in English, creating accessibility barriers for healthcare providers more comfortable with Swahili or other local languages
- **No clinical workflow integration**: Does not integrate with existing electronic health records or patient management systems, requiring manual incorporation of recommendations
- **Unvalidated clinical effectiveness**: Has not undergone extensive field testing with target users, leaving practical effectiveness in real healthcare settings unverified
- **No adaptive learning**: Cannot learn from user interactions or improve responses based on usage patterns and feedback over time

## CONCLUSION & FUTURE WORK

### Conclusion

The MAM-AI initiative represents a transformative approach to addressing the critical information gap that contributes to maternal and newborn mortality in resource-constrained healthcare settings. By developing an offline, AI-powered clinical decision support tool specifically designed for Zanzibar's healthcare realities, we have created a practical solution that places evidence-based maternal care guidance directly

into the hands of frontline nurses and midwives.

Our implementation successfully integrates the Gemma-3n language model with Retrieval Augmented Generation technology, operating entirely offline on standard Android devices. The system's ability to instantly retrieve relevant clinical guidance from a database of expert-curated medical protocols addresses the fundamental "know-do gap" that has historically prevented the translation of medical knowledge into practice at the point of care.

The technical achievements of this project are significant. We successfully implemented an efficient RAG pipeline using Google's AI Edge SDK, and created a user-friendly Flutter interface that prioritizes simplicity and rapid access to information. The entire system operates without internet connectivity, removing the primary barrier to technology adoption in settings with unstable infrastructure. Most importantly, MAM-AI demonstrates that advanced AI technologies can be made accessible and practical for healthcare workers in low-resource settings. By eliminating dependence on internet connectivity, expensive data bundles, and external servers, our solution ensures that every provider—regardless of location or infrastructure constraints—can access the clinical guidance needed to deliver timely, evidence-based care.

This work contributes to the broader goal of healthcare equity by democratizing access to clinical decision support tools. The potential impact extends beyond Zanzibar, offering a scalable model for addressing similar healthcare challenges across Sub-Saharan Africa and other resource-constrained regions worldwide.

## Future Work

The current MAM-AI implementation represents a foundational step toward comprehensive AI-assisted maternal healthcare support, but significant opportunities exist for enhancement across multiple dimensions. The following sections outline key areas for future development that will transform the system from its current search-based functionality into a more sophisticated, clinically impactful tool.

**The fine-tuned model needs to be converted to .task format to enable integration with RAG.** A critical priority for future development is resolving the model conversion challenges encountered during this project. Our next phase will focus on successfully converting the fine-tuned Gemma-3n model to the .tflite format required for Android deployment, addressing the technical obstacles and compatibility issues that prevented integration in the current iteration. Once conversion is achieved, we will implement a comprehensive RAG pipeline that leverages both the fine-tuned model's enhanced capabilities and robust context retrieval mechanisms. This integration is expected to significantly improve response accuracy and contextual relevance within the MAM-AI system. The combined approach of domain-specific fine-tuning and retrieval-augmented generation represents a substantial advancement that will enable more sophisticated, context-aware interactions while maintaining the mobile-optimized performance requirements of our Android application.

**Enhanced Conversational Capabilities.** The current MAM-AI implementation functions as a search-based query system, providing responses to individual healthcare queries

without maintaining conversational context. Future development will focus on transforming the application into a full conversational chatbot capable of multi-turn interactions. This enhancement would enable healthcare providers to engage in dynamic clinical consultations, asking follow-up questions and receiving contextually aware responses that build upon previous interactions within the same case discussion. Such conversational capabilities would better mirror the natural flow of clinical decision-making, where practitioners often need to explore multiple aspects of a case through iterative questioning and refinement of their understanding.

**Search and Chat History Preservation.** Implementation of comprehensive history management features represents a critical next step in the system's evolution. This includes maintaining a searchable log of previous queries and responses to enable providers to quickly reference past consultations, particularly valuable when managing recurring cases or tracking the progression of complex clinical scenarios. Chat session management capabilities would preserve conversational context across multiple interactions for complex clinical cases, allowing practitioners to return to ongoing discussions without losing the thread of their clinical reasoning. Additionally, favorites and bookmarking functionality would allow users to save frequently referenced protocols or particularly relevant responses for quick access, creating personalized libraries of clinical decision support resources tailored to individual practice patterns and commonly encountered scenarios.

**Document Navigation and Source Transparency.** A significant enhancement would involve developing an integrated document viewer that provides unprecedented transparency into the AI's reasoning process and source materials. Source highlighting functionality would automatically highlight the specific sections within source documents that contributed to each response, enabling healthcare providers to quickly identify the exact evidence supporting particular recommendations. Direct document access would enable users to navigate directly to the full source material for comprehensive reading, supporting evidence-based practice by facilitating deeper exploration of clinical guidelines. Citation mapping would provide clickable references that link responses back to specific pages, sections, or protocols in the original guidelines, creating a seamless bridge between AI-generated recommendations and their authoritative sources.

**Dataset Specialization and Expansion.** While the current system utilizes a general medical subset of the MIRIAD dataset, future iterations should incorporate more specialized and contextually relevant training data. Maternal-specific training data would involve fine-tuning on datasets specifically focused on obstetric and maternal health scenarios, improving the system's understanding of the unique challenges and considerations in maternal healthcare. Local clinical protocol integration would incorporate Zanzibar-specific and East African clinical guidelines to ensure cultural and regional relevance, addressing the critical need for healthcare AI systems that reflect local practices, resource constraints, and epidemiological patterns. A continuous learning pipeline would develop mechanisms to regularly update the knowledge base with new clinical evidence and guidelines, ensuring that the system remains current with evolving medical knowledge and best practices.

**Multi-Language Support.** Expansion to support Swahili and other local languages prevalent in Zanzibar would significantly improve accessibility and adoption among healthcare providers who may be more comfortable communicating in their native languages. This enhancement is particularly crucial in settings where English may be a second or third language for healthcare workers, potentially creating barriers to effective utilization of the system. Multi-language support would not only improve usability but also ensure that nuanced medical concepts are accurately conveyed and understood across linguistic boundaries.

**Validation and Clinical Impact Assessment.** Rigorous field testing and clinical validation studies should be conducted to establish the system's effectiveness and safety in real-world clinical environments. These studies should measure the impact of MAM-AI on clinical decision-making and patient outcomes, providing quantitative evidence of the system's contribution to improved maternal and newborn health. User satisfaction and adoption rate assessments among healthcare providers would identify barriers to implementation and areas for user experience improvement. Validation of the accuracy and clinical relevance of AI-generated recommendations would ensure that the system meets the high standards required for clinical decision support tools, building trust among healthcare providers and establishing the foundation for broader deployment.

These future developments will build upon the solid foundation established by the current MAM-AI system, ultimately creating a more comprehensive, user-friendly, and clinically impactful tool for improving maternal and newborn health outcomes in resource-constrained settings. The evolution from a basic search interface to a sophisticated conversational clinical assistant represents a significant opportunity to advance AI-supported healthcare in contexts where such innovations can have the greatest impact on global health equity.

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